

Classification of exceptional nodal topologies protected by $\mathcal{PT}$ symmetry

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Exceptional degeneracies, at which both eigenvalues and eigenvectors coalesce, and parity-time ($\mathcal{PT}$) symmetry, reflecting balanced gain and loss in photonic systems, are paramount concepts in non-Hermitian systems. We here complete the topological classification of exceptional nodal degeneracies protected by $\mathcal{PT}$ symmetry in up to three dimensions and provide simple example models whose exceptional nodal topologies include previously overlooked possibilities such as second-order knotted surfaces of arbitrary genus, third-order knots, and fourth-order points.

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I. INTRODUCTION

Parity-time ($\mathcal{PT}$) symmetric non-Hermitian (NH) descriptions were originally suggested as a fundamental amendment to the standard quantum mechanics motivated by their capacity of having real spectra [1,2]. By now, NH models are instead recognized as effective descriptions with a remarkably wide range of applications [3], and $\mathcal{PT}$ symmetry is a ubiquitous feature of photonic experiments where it reflects a balance between gain and loss [4]. Exceptional point (EP) degeneracies at which both eigenvalues and eigenvectors coalesce are central and uniquely NH features associated with a rich phenomenology in these systems [5]. Recently, the merger between topological [6,7] and NH physics has led to the burgeoning field of NH topological systems [8].

The topological properties of NH nodal band structures, consisting of EPs, are central topics within this field [8]. EPs are topologically distinguished from Hermitian degeneracies in that they are generic in the bulk of two-dimensional (2D) systems [9–12], giving closed (potentially knotted) lines of EPs in three dimensions (3D) [13–19]. In general, EPs of order n (n EPs) require the tuning of $2n - 2$ real parameters, resulting in their absence for $n \geq 3$ in systems of 3D or lower. The inclusion of discrete symmetries can, however, relax this constraint and make symmetry-protected 2EPs generic already in one dimension (1D), while rings and surfaces of 2EPs are stabilized in 2D and 3D, respectively [20–25]. These EPs are associated with open Fermi arc degeneracies in the real part of the spectrum with telltale signatures in experiments [9]. Very recently, it was shown that physically relevant symmetries can also stabilize higher-order EPs, and explicit

examples of models with pointlike 3EPs in 2D were explored [26,27].

Here, we show that $\mathcal{PT}$ symmetry protects a remarkably rich family of nodal EP topologies in 3D. By explicit construction, we below provide a classification thereof, including possibilities such as knotted surfaces of arbitrary genus of 2EPs, knotted lines of 3EPs, and pointlike 4EPs, cf. Fig. 1 and Table I. All structures are generic and stable in the sense that they are indeed protected by $\mathcal{PT}$ symmetry; small perturbations can destroy the EP topologies only if they break $\mathcal{PT}$. This points to their experimental relevance mainly in photonic systems where the $\mathcal{PT}$ can be realized with great precision and system parameters may be tuned with excellent control [4].

II. $\mathcal{PT}$ AND CODIMENSION OF NEPS

The codimension of n EPs is related to the properties of the corresponding characteristic polynomial $\text{Ch}(\lambda)$. Imposing $\mathcal{PT}$ symmetry yields the following constraints on $\text{Ch}(\lambda)$ [26]:

$$\text{Ch}_{\mathcal{PT}}(\lambda) = \det(H_{\mathcal{PT}} - \lambda) = \det(H_{\mathcal{PT}}^* - \lambda), \quad (1)$$

where $*$ denotes complex conjugation. Explicitly, Eq. (1) reads

$$\text{Ch}_{\mathcal{PT}}(\lambda) = \sum_{j=0}^n a_j \lambda^j = \sum_{j=0}^n a_j^* \lambda^j. \quad (2)$$

Consequently, a_j are real-valued functions, indicating that $\mathcal{PT}$ reduces the number of real constraints for n EPs to $n - 1$, thus requiring the tuning of at least $n - 1$ real parameters. Below, we show by explicit construction that these constraints have real-valued solutions in 3D, going beyond previous works such as Ref. [26]. The general argument confirms that a further decrease of the codimension of the EPs are not stabilized by $\mathcal{PT}$ and therefore relies on fine-tuning. Consequently, our findings constitute a classification of EP topologies protected by $\mathcal{PT}$ symmetry in 3D since the exceptional structures map over to embeddings of the respective dimension, ensuring that all corresponding topologies can be obtained [28–30].

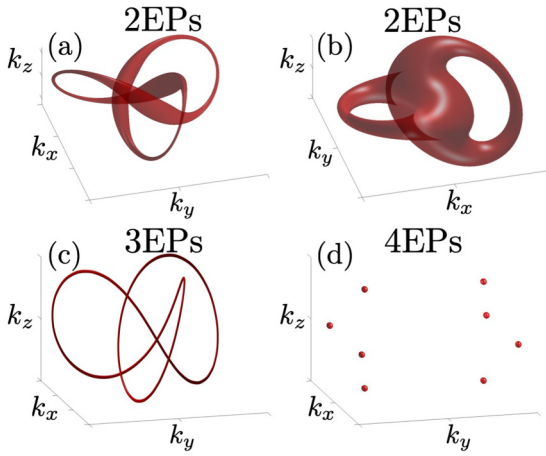


FIG. 1. $\mathcal{PT}$ symmetry-protected exceptional nodal degeneracies in three-dimensional (3D) systems. In (a) and (b), second-order exceptional surfaces of the Hamiltonian in Eq. (4) are displayed. (a) An inflated exceptional trefoil knot. (b) An inflated Hopf link. The surfaces are obtained by adding a symmetry-preserving perturbation to systems hosting ordinary nodal knot degeneracies. The surfaces (a) and (b) are of genus 1 and 2, respectively, and higher genus surfaces can be obtained in a systematic manner. (c) An exceptional trefoil knot of order three is shown, which generically occurs in $\mathcal{PT}$ -symmetric models with three or more bands, e.g., Eq. (5). Lastly, exceptional points of order four are displayed in (d). Together, these exceptional structures provide a complete classification of exceptional nodal topologies protected by $\mathcal{PT}$ symmetry in 3D, cf. Table I.

Throughout this Letter, we use a representation such that $\mathcal{PT}$ symmetry amounts to complex conjugation, meaning that matrix representations of $\mathcal{PT}$ -symmetric Hamiltonians consist of purely real entries and $(\mathcal{PT})^2 = 1$.

III. EXCEPTIONAL NODAL SURFACES

$\mathcal{PT}$ symmetry preserving anti-Hermitian terms split ordinary Hermitian nodal points to exceptional rings in two-band systems [20]. Generalizing this reasoning to higher dimensions, nodal lines split to exceptional surfaces when exposed to similar perturbations. To demonstrate this, we first show that $\mathcal{PT}$ symmetry-protected nodal torus knots are split into exceptional knotted tori. Consider the following $\mathcal{PT}$ -symmetric Hermitian Hamiltonian [31]:

$$H_{(p,q)} = \text{Re}[\zeta^p(k_x, k_y, k_z) + \xi^q(k_x, k_y, k_z)]\sigma^x + \text{Im}[\zeta^p(k_x, k_y, k_z) + \xi^q(k_x, k_y, k_z)]\sigma^z, \quad (3)$$

where a σ^y term is absent due to the symmetry. Here, $\zeta(k_x, k_y, k_z) = k_x + ik_y$, $\xi = k_z + i[M - (k_x^2 + k_y^2 + k_z^2)]$, $M \in \mathbb{R}$, σ^i the Pauli matrices, and k_i the lattice momentum components. Here, p and q are positive integers, and the corresponding nodal points form torus knots when p and q are relatively prime, and torus links otherwise. By adding a constant symmetry-preserving anti-Hermitian term, resulting in the following Hamiltonian:

$$H = H_{(p,q)} + i\delta\sigma^y, \quad (4)$$

for $\delta \in \mathbb{R}$, the nodal knots are inflated to form second-order exceptional surfaces of knotted tori, displayed in Figs. 1(a),

TABLE I. The impact of $\mathcal{PT}$ symmetry on the exceptional nodal topologies in systems of physically relevant dimensions.

Dimension	Exceptional nodal topologies	
	Without $\mathcal{PT}$	With $\mathcal{PT}$
$D = 1$	—	2EPs
$D = 2$	2EPs	Lines of 2EPs 3EPs
$D = 3$	Knots of 2EPs	Surfaces of 2EPs Knots of 3EPs 4EPs

2(a), and 2(b). This construction is, however, not restricted to torus knots but is also applicable for the hyperbolic Turk's head knots by following the construction in Ref. [17]. Details can be found in the online Supplemental Material [32].

Despite being topologically different, in the sense that the different knotted tori cannot be continuously deformed into each other, they are all different examples of embeddings of the regular torus into $\mathbb{R}^3$. Thus, the genus of a knotted torus is always 1—the genus is independent of the embedding. Since embeddings of 2D surfaces in 3D spaces are completely classified by the corresponding genus, we expect higher genus exceptional surfaces to appear as well. To this end, let us again consider the Hamiltonian in Eq. (4), and let $p = q = n$. The exceptional structure for H consists of n individually linked tori. When δ becomes large enough, the tori are merged together close to the origin of momentum space, and the exceptional structure attains the form of a sphere with n handles, i.e., a genus n surface. This is explicitly shown in Figs. 1(b) and 2(c) for $n = 2$ and 3, respectively, while higher genus examples are provided in the Supplemental Material [32]. Thus, it is possible to obtain second-order exceptional surfaces with any genus. These surfaces are furthermore generic and stable, in the sense that they are protected by $\mathcal{PT}$ symmetry.

IV. KNOTTED EXCEPTIONAL LINES OF ORDER THREE

The plethora of exceptional nodal topologies protected by $\mathcal{PT}$ symmetry does not only contain second-order surfaces but also higher-order degeneracies. Below, building on the recent examples on 3EPs (points) in 2D [26,27], we exemplify this by explicit construction of a three-band model hosting knotted exceptional lines of order three. Consider the Hamiltonian given by

$$H_3 = \begin{pmatrix} f_2 & 0 & \Lambda \\ \alpha & 0 & \beta \\ f_2 & f_1 & -f_2 \end{pmatrix}, \quad (5)$$

where f_1 and f_2 are continuously differentiable functions of the lattice momentum components, and $\alpha, \beta, \Lambda \in \mathbb{R}$. The corresponding characteristic equation determining the eigenvalues reads

$$f_1(\alpha \Lambda - \beta f_2) + \lambda(f_2^2 + \Lambda f_2 + \beta f_1) - \lambda^3 = 0. \quad (6)$$

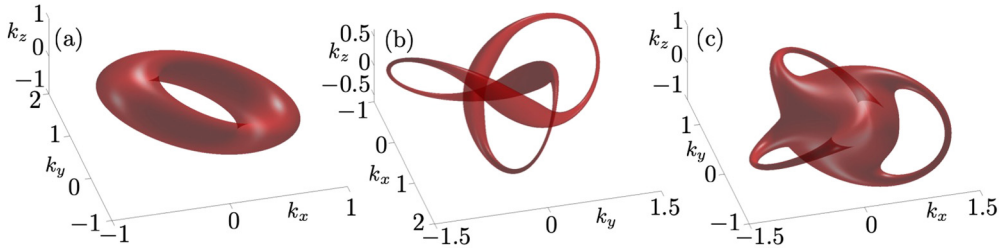


FIG. 2. Second-order exceptional surfaces of the Hamiltonian in Eq. (4). In all panels, $M = 0.5$. In (a), $p = q = 1$ and $\delta = 0.4$, resulting in an exceptional torus. In (b), $p = 3$, $q = 2$, and $\delta = 0.15$, resulting in an inflated exceptional trefoil knot. Despite appearing topologically different, both (a) and (b) display genus 1 surfaces since the inflated trefoil knot is an embedding of the torus. In (c), $p = q = 3$ and $\delta = 0.4$, resulting in a genus 3 surface. In this controlled manner, exceptional nodal surfaces of any topology can be achieved by choosing $p = q = n$.

To have third-order exceptional structures, all eigenvalues have to coalesce. This happens exactly when

$$f_1(\alpha \Lambda - \beta f_2) = 0, \quad (7)$$

$$(f_2^2 + \Lambda f_2 + \beta f_1) = 0. \quad (8)$$

When $|\Lambda|$ is sufficiently large, the only solution becomes $f_1 = f_2 = 0$. Recalling the interpretation of a knot as the intersection of two implicitly defined surfaces, f_1 and f_2 can be defined such that their common zeros resemble any knot [29,30]. By choosing

$$f_1 = \text{Re}[\zeta^p(k_x, k_y, k_z) + \xi^q(k_x, k_y, k_z)], \quad (9)$$

$$f_2 = \text{Im}[\zeta^p(k_x, k_y, k_z) + \xi^q(k_x, k_y, k_z)], \quad (10)$$

with ζ and ξ as above, the exceptional lines are torus knots [28], while an exceptional hyperbolic figure-eight knot is obtained when

$$f_1 = (k_y^2 - k_z^2)\epsilon^2 + R(8R^2 - 2\epsilon^2), \quad (11)$$

$$f_2 = 2\sqrt{2}Rk_yk_z + k_x(8R^2 - 2\epsilon^2), \quad (12)$$

where $R^2 = \epsilon^2 - (k_x^2 + k_y^2 + k_z^2)$ [33]. In Fig. 3, a sample of illustrations is provided, including examples from both of the cases above. The topology of the knots can be determined by computing knot invariant polynomials, such as the Alexander polynomial [17,34]. In the Supplemental Material [32], additional examples, including hyperbolic Turk's head knots, are provided.

To emphasize that the knotted exceptional structures are indeed of order three, let us consider the eigenvectors, which

are given by

$$V_i = \left[-\frac{\Lambda}{f_2 - E_i}, -\frac{-\beta f_2 + \alpha \Lambda + \beta E_i}{(f_2 - E_i)}, 1 \right]^T, \quad (13)$$

where E_i denotes the corresponding eigenvalue, and $i = 1, 2, 3$. Taking the difference $V_i - V_j$ for $i \neq j$, and letting $E_i \rightarrow E_j$, the eigenvectors indeed coalesce when the eigenvalues do, and hence, the degeneracies at $E_i = 0$ are exceptional of order three. They are furthermore stable toward symmetry-preserving perturbations, which follows by the stability of the models in Refs. [16,17].

V. 4EPS IN PERTURBED DIRAC SEMIMETALS

Finally, we demonstrate the existence of 4EPs in $\mathcal{PT}$ -symmetric four-band systems. Let us start with the Hamiltonian of a real Dirac semimetal:

$$H_D = k_x \Gamma^1 + k_y \Gamma^2 + k_z \Gamma^3, \quad (14)$$

where $\Gamma^1 = \sigma_1 \otimes \tau_0$, $\Gamma^2 = \sigma_2 \otimes \tau_2$, and $\Gamma^3 = \sigma_3 \otimes \tau_0$, together with $\Gamma^4 = \sigma_2 \otimes \tau_1$ and $\Gamma^5 = \sigma_2 \otimes \tau_3$ constitute a representation of the Clifford algebra. Here, σ_i and τ_i denote the Pauli matrices. Anticommuting $\Gamma^1, \dots, \Gamma^4$ provides a complete basis of 4×4 matrices. Adding an anti-Hermitian term of the form $H_{AH} = i\delta_4 \Gamma^4 + \delta_6 \Gamma^6 + k_x \delta_{12} \Gamma^{12}$, to obtain

$$H = H_D + H_{AH}, \quad (15)$$

with $\Gamma^6 = \Gamma^1 \Gamma^2$, $\Gamma^{12} = \Gamma^1 \Gamma^2 \Gamma^3$ and $\delta_4, \delta_6, \delta_{12} \in \mathbb{R}$, splits the Dirac node at energy $E = 0$ into eight 4EPs, all located at $E = 0$. Here, $A^{[\mu_1 \dots \mu_n]} = \frac{1}{n!} \epsilon_{\mu_1 \dots \mu_n} A^{\mu_1} \dots A^{\mu_n}$, with $\epsilon_{\mu_1 \dots \mu_n}$ the Levi-Civita symbol. This can be seen by studying the characteristic polynomial, which attains the form of

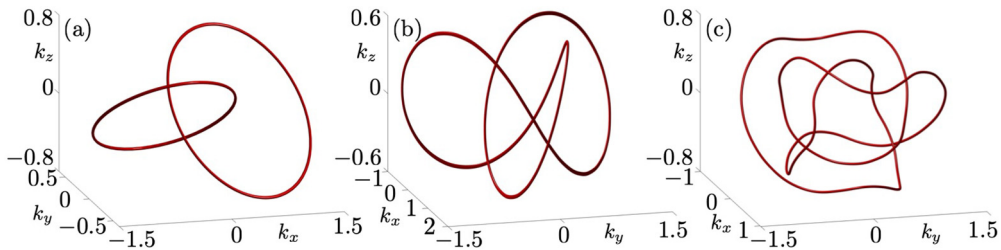


FIG. 3. Knotted and linked exceptional lines of order three appearing as triple eigenvalue degeneracies in the model described by Eq. (5). For all panels, $\alpha = 1$, $\beta = -1$, and $\Lambda = 100$. (a) An exceptional Hopf link, which corresponds to $p = q = 2$ in Eqs. (9) and (10). (b) An exceptional trefoil knot, for which $p = 3$ and $q = 2$. In both cases, $M = 0.5$. The exceptional knot in (c) is a figure-eight knot, appearing when choosing f_1 and f_2 as in Eqs. (11) and (12), with $\epsilon = 1$.

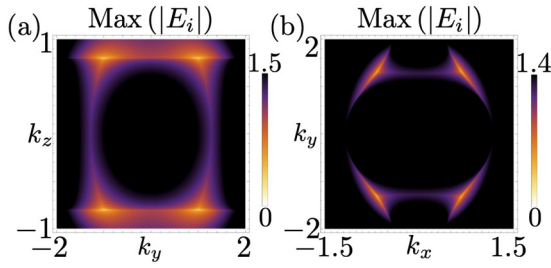


FIG. 4. 4EPs appearing in the system described by the Hamiltonian in Eq. (15). In both panels, $\delta_4 = 0.8$, $\delta_6 = 1$, and $\delta_{12} = 1.1$. In (a), $k_x = 0$, and the 4EPs occur at $(k_y, k_z) = (\pm_1 \delta_6, \pm_2 \delta_4)$, where $\pm_1$ and $\pm_2$ are independent of each other. In (b), $k_z = 0$, and the 4EPs occur at $(k_x, k_y) = (\pm_1 0.669048, \pm_2 1.31681)$. Since 4EPs necessarily occur at $E = 0$, they correspond exactly to when the maximum of the absolute value of all eigenvalues vanish simultaneously. These points are furthermore stable toward any small but finite perturbation and are thus generic in systems protected by $\mathcal{PT}$ symmetry.

a depressed quartic. If such polynomials have fourfold degenerate solutions, they are necessarily located at $E = 0$. In Fig. 4, the maximum of the absolute values of all eigenvalues $E_1, \dots, E_4$ are displayed as a density plot in the appropriate planes in momentum space. The 4EPs occur exactly when the maximum value is zero since then all eigenvalues necessarily coalesce at $E = 0$. The points are indeed exceptional since the dimensions of the eigenspaces at those points are one, indicating that the eigenvalues $E_1 = E_2 = E_3 = E_4 = 0$ share one eigenvector.

VI. DISCUSSION

In this Letter, we provide a classification of exceptional nodal topologies protected by $\mathcal{PT}$ symmetry in 3D. Through

explicit construction, we demonstrate the existence of several hitherto overlooked types of exceptional nodal degeneracies: knotted surfaces of arbitrary genus of 2EPs, knotted lines of 3EPs, and 4EPs, thus extending the already rich plethora of NH topological phases [8,20–23,35–43]. We note that higher-order EPs require tuning of more parameters than allowed by dimensional constraints, making the list of exceptional nodal topologies complete for physically relevant systems if additional tuning variables are not considered.

We emphasize again that our findings rely only on the presence of $\mathcal{PT}$ symmetry. Being arguably the most widely implemented symmetry in photonic experimental setups makes them highly relevant in optics [4]. It should be possible to experimentally simulate the models we present using single-photon interferometry as in Ref. [44]. We furthermore note that 2EPs, together with their bulk Fermi arcs, as well as the more exotic knotted lines of 2EPs and their concomitant Fermi-Seifert surfaces were both experimentally realized and observed shortly after being theoretically predicted [9,44]. This suggests that topological phases of higher-order EPs are within experimental reach in several platforms and encourages the search for tailor-made setups where they can be realized.

Note added in proof. An independent work shows how symmetry-protected 4EPs may naturally be realized in three-dimensional correlated many-body systems [45].

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